

Ques. WHICH OF THE EASILY-AVAILABLE PCB LAYOUT DESIGN SOFTWARE CAN DIRECTLY CREATE ISLANDS AND CONNECT/CREATE TRACKS MANUALLY?

S.M.B. Anand

Ans. There are many PCB layout designing software available on the Internet, of which many are free and easy to use, including KiCad, gEDA, PCB123, PCBArtist and EasyEDA. These are easy-to-use for creating islands and tracks, among others. PCB components can be viewed in 3D

LED lighting requires direct current (DC) rather than an AC power supply to eliminate flickering issues. However, flickering is common in almost all types of LED lamps. Let us see some examples of flickering lights in addition to detecting the symptoms and steps to fix these.

Some possible reasons for flickering are loose wiring and other faulty connections, non-compatible dimmer switch, power supply issue in the driver circuit (if there is no dimmer) and fluctuations in the power grid. You need to choose a high-quality LED power supply to reduce visible flicker.

The easiest way for troubleshooting is by swapping the faulty lamp with another one, if there are several identical LED lamps installed in the house. Then, check if flickering occurs with the other lamp also.

If lights are producing a regular, rhythmic flicker, the LEDs are likely to be drawing more power than the driver circuit allows. If the lights flicker for a short time after switching on, it is either because of their poor quality or these are about to become faulty. Also, components used in the integrated power supply may not be of good quality.

If the LED light works perfectly for a long time after initial flickering, a faulty capacitor inside the driver circuit could be responsible. Check for any loose contact of components in the circuit.

The last option is to replace with a new LED light. However, an electrician might also help you find the exact cause and fix the issue. We advise that you consult a licensed electrician to properly diagnose the issue.

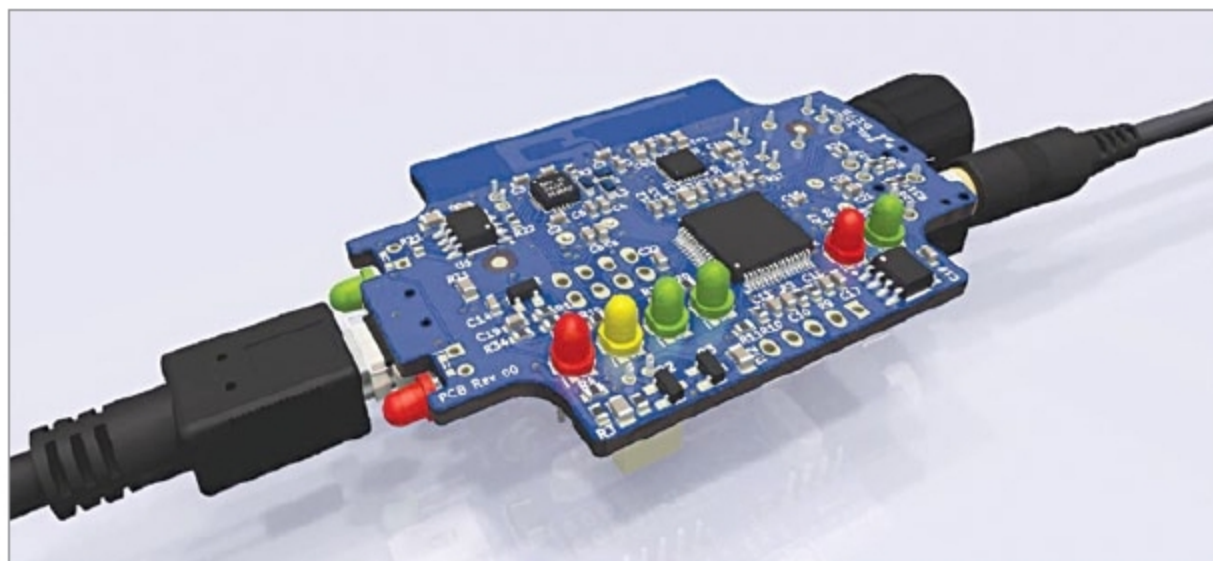


Fig. 1: 3D view of KiCad designed PCB

with some of these software. Another popular option is Eagle, but it is not free. All the above-mentioned software are easily available for download from the Internet.

Q2. WHY DO LED LAMPS FLICKER WHEN SWITCHED ON?

Suresh

A2. Flickering is common in compact lights including retrofit LED lamps (Fig. 2) and compact fluorescent light tubes. LED lights flicker in many varying styles including initial flickering when these are switched on. Reasons for flickering could either be a simple issue or a more intricate problem. Flickering lights can be irritating.



Fig. 2: Typical retrofit LED lamps (Credit: luminwiz.com)

Q3. HOW CAN I HAVE MULTIPLE INDICATIONS USING RED, AMBER AND GREEN LEDs FOR DIFFERENT STATUSES IN A CIRCUIT?

Mohsen Salimy

A3. You can have multiple LED indications using a single RGB LED, different LEDs with different colours or a string of LEDs with different colours. A single RGB LED for multiple indications is the most common and easiest one. You may please see "Multipurpose Indicator Using Single RGB LED" DIY article on EFY website <https://electronicsforu.com>

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender.